Q. 74. If $D=\begin{bmatrix} 0 & a\alpha^2 & a\beta^2 \\ b\alpha+c & 0 & a\gamma^2 \\ b\beta+c & (b\gamma+c) & 0 \end{bmatrix}$ is a skew symmetric matrix (where α, β, γ are distinct) and the value of $\begin{vmatrix} (a+1)^2 & (1-a) & (2-c) \\ (3+c) & (b+2)^2 & (b+1)^2 \\ (3-b)^2 & b^2 & (c+3) \end{vmatrix}$ is λ, then the value of $|10\lambda|$ is.....

Q. 75. Let $|z| = |z-3| = |z-4i|$, then the value $|2z|$ is.....

Answers

Physics

Q. No.	Answer	Topic's name	Q. No.	Answer	Topic's name
1	(3)	Experimental Physics	14	(3)	Nuclear Fission
2	(3)	Kinetic Theory of Gases	15	(1)	Unit and Measurement
3	(1)	Heat Transfer	16	(1)	Laws of Motion
4	(1)	Sound Wave	17	(2)	Electric Circuit, Electrical Power
5	(4)	Self Inductance	18	(4)	Series LCR Circuit
6	(1)	Wave Optics	19	(2)	Atoms
7	(4)	Torque on a Current-carrying loop placed in a Magnetic field	20	(1)	Mechanical properties of Fluids
8	(3)	Motion in a Vertical Circle	21	[5.5]	Electronic Devices
9	(3)	Young's Double Slit Experiment	22	[205.00]	Thermodynamics
10	(3)	Logic Gates	23	[2.00]	Conservation of Mechanical energy
11	(3)	Moment of Inertia	24	[1.20]	Electrostatic Potential
12	(3)	Gravitation	25	[4.68]	Mechanical properties of Solids
13	(2)	Stopping Potential, X - Rays			

Chemistry

Q. No.	Answer	Topic's name	Q. No.	Answer	Topic's name
26	(1)	Periodic Table	39	(3)	Hydrocarbon
27	(4)	Chemical Bonding	40	(1)	Halogen Dervitative
28	(1)	Salt Analysis	41	(3)	P-block
29	(3)	Thermodynamics	42	(3)	Alcohol, Ether and Phenol
30	(2)	Ionic Equilibrium	43	(4)	Aldehyde and Ketone
31	(3)	Ionic Equilibrium	44	(1)	Carboxylic Acid
32	(1)	IUPAC	45	(3)	Nitrogen Compound
33	(1)	Isomerism	46	[15.00]	Chemical Kinetics
34	(1)	Hydrocarbon	47	[0.34]	Electrochemistry
35	(2)	Ketones	48	[2]	Alcohol
36	(2)	Lanthanoids	49	[57]	Thermodynamics
37	(3)	Solutions	50	[8]	Coordinaion Chemistry
38	(2)	General Organic Chemistry			

Mathematics

Q. No.	Answer	Topic's name	Q. No.	Answer	Topic's name
51	(2)	Set Theory, Relation	64	(4)	Hyperbola
52	(2)	Vector	65	(3)	Quadratic Equations
53	(3)	Indefinite Integration	66	(2)	Determinants
54	(2)	Definite Integration	67	(1)	Arithmetic Progression
55	(1)	Area under the curve	68	(2)	Binomial Theorem
56	(3)	Indefinite integration	69	(1)	Permutations and Combinations
57	(2)	Application of Derivatives – Maxima & Minima	70	(3)	Point and Straight Line
58	(3)	Differentiation	71	[19]	Three Dimensional Geometry
59	(3)	Functions	72	[17]	Probability
60	(2)	Limits	73	[78]	Definite Integral
61	(1)	Circles	74	[600]	Determinants
62	(1)	Statistics	75	[5.00]	Complex Numbers
63	(3)	Three Dimensional Geometry			

ANSWERS WITH EXPLANATION

Physics

1. Option (3) is correct.

$$\text{The least count} = \frac{1}{50}$$

So, $$\text{Vernier constant} = \frac{1}{50} \times 0.5 \text{ mm}$$

$$= 0.01 \text{ mm}$$

2. Option (3) is correct.

$$P \propto m$$

Since m is increased by a factor of $\frac{4}{3}$,

P will increase by a factor of $\frac{4}{3}$.

$$\therefore \quad \text{New pressure} = \frac{3}{2} \times 76 \text{ cm of Hg}$$

$$= 114 \text{ cm of Hg}$$

3. Option (1) is correct.

$$H = \text{rate of heat flow}$$

$$= \frac{900}{\frac{l_i}{K_i A} + \frac{l_0}{K_0 A}}$$

Now $$1000 - \theta = \frac{H l_i}{K_i A}$$

or $$\theta = 1000 - \left[\frac{900}{\frac{l_i}{K_i A} + \frac{l_0}{K_0 A}}\right] \frac{l_i}{K_i A}$$

$$= 1000 - \frac{900}{1 + \frac{l_0}{K_0}\frac{K_i}{l_i}}$$

Now, we can see that θ can be decreased by increasing thermal conductivity of outer layer (K_0) and thickness of inner layer (l_i).

4. Option (1) is correct.

$$e = \frac{l_2 - 3l_1}{2}$$

$$e = \frac{70.2 - 3 \times 22.7}{2} = \frac{70.2 - 68.1}{2}$$

$$= \frac{2.1}{2} = 1.05 \text{ cm}$$

5. Option (4) is correct.

Self-inductance of coil:

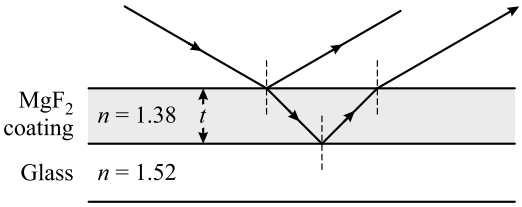
$$L = \frac{N\phi}{I}$$

$$= \frac{1000 \times 4 \times 10^{-3}}{4} = 1 \text{ H}$$

$$= 1 \text{ H}$$

6. Option (1) is correct.

$$2t = \left[\frac{2m+1}{2}\right]\frac{\lambda}{n}$$



$$t_{min} = \frac{\lambda}{4n} \qquad [\text{by putting } m = 0]$$

$$= \frac{5.5 \times 10^{-7}}{4 \times 1.38} = 99.6 \text{ nm}$$

7. Option (4) is correct.

$$\vec{M} = i\vec{A}$$

$$= 5 \times (0.2) \times (0.1)\left(-\hat{i}\right)$$

$$= 0.1\left(-\hat{i}\right)$$

$$\vec{\tau} = \vec{M} \times \vec{B}$$

$$= 0.1\left(-\hat{i}\right) \times (2 \times 10^{-3})\left(\hat{j}\right)$$

$$= 2 \times 10^{-4}\left(-\hat{k}\right) \text{ N} - m$$

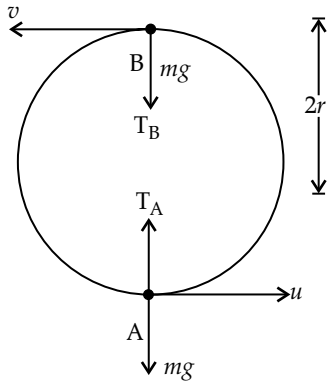
8. Option (3) is correct.

At lowest point,

$$T_A - mg = \frac{mu^2}{r} \qquad ...(1)$$

At highest point,

$$T_B + mg = \frac{mv^2}{r} \qquad ...(2)$$



Gain in PE from from A to B

$$= 2\,mgr = \text{loss in KE}$$

$$\Rightarrow \quad 2mgr = \frac{1}{2}m\ (u^2 - v^2)$$

$$\Rightarrow \quad u^2 - v^2 = 4\ gr \qquad ...(3)$$

from (1), (2), and (3)

$$T_A - T_B - 2mg = \frac{m}{r}\ (u^2 - v^2)$$

$$T_A - T_B = 2mg + \frac{m}{r} \times 4gr = 6\ mg$$

9. Option (3) is correct.

$$\frac{I_{max}}{I_{min}} = \frac{9}{1} = \frac{(a_1 + a_2)^2}{(a_1 - a_2)^2}$$

$$\Rightarrow \quad \frac{a_1 + a_2}{a_1 - a_2} = \frac{3}{1}$$

$$\Rightarrow \quad a_1 + a_2 = 3a_1 - 3a_2$$

$$-4a_2 = -2a_1$$

$$\frac{a_1}{a_2} = \frac{2}{1}$$

10. Option (3) is correct.

$$Y' = \overline{A + B}$$

$$Y = \overline{Y'}$$

$$Y = \overline{\overline{A + B}}$$

$Y = A + B$ which is an OR-gate

11. Option (3) is correct.

M.I. of the uniform circular disc about its diameter = I

M.I. of the disc about its axis = $\frac{1}{2}mr^2$

According to the Theorem of perpendicular axis,

$$I_z = I_x + I_y$$

M.I. of the disc about its axis,

$$I_z = I + I = 2I$$

Thus, $2I = \frac{1}{2}mr^2$

$\therefore \quad mr^2 = 4I$

As per the parallel axes theorem, moment of inertia of disc about the perpendicular axis passing through its rim:

$$= \frac{1}{2}mr^2 + mr^2$$

$$= 2I + 4I = 6I$$

12. Option (3) is correct.

The Equivalence Principle, which asserts that the mass responsible for an object's resistance to acceleration (inertial mass) is identical to the mass that determines the strength of its gravitational attraction (gravitational mass). So, the ratio of the inertial mass to gravitational mass is equal to 1.

13. Option (2) is correct.

Let E_K, E_L, E_M, E_N be the binding energies of K, L, M and N shell. Let E_P be the energy of the incident photon, then

$$E_P - E_K = 24 \text{ keV} \qquad ...(1)$$

$$E_P - E_L = 100 \text{ keV} \qquad ...(2)$$

$$E_P - E_M = 110 \text{ keV} \qquad ...(3)$$

From equation (1) and (2)

$$E(K_\alpha) = E_K - E_L$$

$$= 100 - 24 = 76 \text{ keV}$$

14. Option (3) is correct.

$$^{200}X \longrightarrow {}^{120}Y + {}^{80}Z$$

Energy released = Nuclear energy of products – Nuclear energy of reactants

Energy released = $80 \times 7 + 120 \times 8 - 200 \times 6.5$

= 220 MeV

15. Option (1) is correct.

Let the time period depend on mass (m), amplitude (a) and constant (k) as:

$$T \propto m^\alpha a^\beta k^\gamma$$

$$[M^\circ L^\circ T] \propto M^\alpha L^\beta \left[\frac{ML^2T^{-2}}{L^3}\right]^\gamma$$

$$[M^\circ L^\circ T] \propto M^{\alpha+\gamma}\ L^{\beta-\gamma} T^{-2\gamma}$$

Equating dimensions on both sides

$$-2\gamma = 1$$

$$\Rightarrow \quad \gamma = -\frac{1}{2}$$

$$\alpha + \gamma = 0$$

$$\Rightarrow \quad \alpha = -\gamma$$

$$\alpha = \frac{1}{2}$$

$$\beta - \gamma = 0$$

$$\Rightarrow \quad \beta = \gamma$$

$$\beta = -\frac{1}{2}$$

$$T \propto m^{1/2}\ a^{-1/2}\ k^{-1/2}$$

$$T \propto \sqrt{\frac{m}{ak}}$$

$$T \propto \frac{1}{\sqrt{a}}$$

16. Option (1) is correct.

$$F - 2T = 6a$$

and $T = 4 \times 2a$

$\therefore \quad F - 16\ a = 6a$

$$\Rightarrow \quad a = \frac{F}{22} \quad \Rightarrow \quad a = 1 \text{ m/s}^2$$

$$\therefore \quad a_{CM} = \frac{6 \times 1 + 4 \times 2}{10} = 1.4 \text{ m/s}^2$$

17. Option (2) is correct.

Maximum power transfer theorem:

Internal Resistance = External Resistance

$$2 = (4 + 2) \,||\, x.$$

$$\Rightarrow \quad 2 = \frac{6x}{6+x}$$

$$\Rightarrow \quad 12 + 2x = 6x$$

$$\therefore \quad x = 3\ \Omega$$

18. Option (4) is correct.

It is given in the problem that

$$V_R = 8V$$

$$V_C = 12V$$

and $V_L = 6V$

In a serises LCR circuit, the applied voltage can be given as

$$V = \sqrt{V_R^2 + (V_C - V_L)^2}$$

Now $V = \sqrt{8^2 + (12-6)^2}$

$$\Rightarrow \quad V = \sqrt{64+36}$$

$$\Rightarrow \quad V = \sqrt{100}$$

$$\Rightarrow \quad V = 10V$$

Now $\cos\phi = \frac{R}{Z}$

$$\Rightarrow \quad \cos\phi = \frac{IR}{IZ}$$

$$\Rightarrow \quad \cos\phi = \frac{V_R}{V}$$

Now putting the values of V_R and V in the equation,

$$\Rightarrow \quad \cos\phi = \frac{8}{10}$$

$$\Rightarrow \quad \cos\phi = \frac{4}{5}$$

Also, $V_L = V_C$

So, We get $I(X_L) < (X_C)$

$$\Rightarrow \quad X_L = X_C$$

Hence, the correct answer is option (4).

19. Option (2) is correct.

$$F = \frac{kq^2}{R^2} = \frac{mv^2}{R}$$

$$\Rightarrow \quad v^2 \propto \frac{1}{R}$$

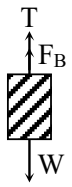
$$\left(\frac{2\pi R}{T}\right)^2 \propto \frac{1}{R}$$

$$\frac{R^2}{T^2} \propto \frac{1}{R}$$

$$R^3 \propto T^2$$

or $T \propto R^{3/2}$.

20. Option (1) is correct.



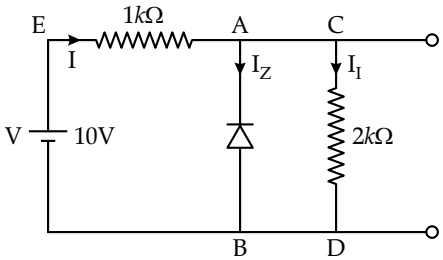
$$T + F_B - W = 0$$

$$W = T + F_B$$

$$= 2.54 \times 9.8 + 10^3 \times 2.197 \times 10^{-3} \times 9.8$$

$$W = 46.43\ N$$

21. Correct answer is [5.5].



Given, $V_Z = 3V$

Let Potential at B = 0V

Potential at E = 10V

Now, $V_C = V_A = 3V$

From the figure

$$I_Z + I_1 = I$$

$$I = \frac{10-3}{1,000} = \frac{7}{1,000}A$$

$$I_1 = \frac{3}{2,000}A$$

Therefore, $I_Z = \frac{7-1.5}{1,000} = 5.5\ mA$

22. Correct answer is [205.00].

$$\text{Work done } W = \frac{1}{2}(40-20)\,(2-1)$$

$$= 10\ \text{Atm} - \text{L} = 1000\ \text{J}.$$

In cyclic process,

$$\Delta W = 0.$$

$\therefore$ As per the first law of thermodynamics,

$$\Delta Q = W = 1000\ \text{J}.$$

$\therefore$ Heat rejected = 1205 – 1000 = 205 J.

23. Correct answer is [2.00].

From conservation of momentum

$$P_i = P_f$$

$$1(2) + 1(0) = (1+1)v$$

$$2 = 2v$$

$$v = 1\ \text{m/s}$$

Now, the initial KE,

$$K_i = \frac{1}{2}(1)(2)^2 = 2J$$

and the final KE,

$$K_f = \frac{1}{2}(2)(1)^2 = 1J$$

$$\text{Loss in K.E.} = (2-1)\text{J} = 1\text{J}$$
$$= \text{Gain in PE}$$
$$1\text{J} = \frac{1}{2}kx^2$$
$$1 = \frac{1}{2}\times 50\times x^2$$
$$x = \frac{1}{5} = 0.2 = \frac{2}{10}\text{m}$$
$$x = 2.$$

24. Correct answers is [1.20].

$$\text{W by electric field} = -5\times 10^{-8}\text{ J}$$
$$\text{W against electric field} = +5\times 10^{-8}\text{ J}$$
$$2\times 10^{-6}\,[2\times a\times (0.1)^2 - 0] = +5\times 10^{-5}\text{ J}$$
$$4\times 10^{-6}\times a\times \frac{1}{100} = 5\times 10^{-8}$$
$$4\times 10^{-8}\times a = 5\times 10^{-8}$$
$$a = \frac{5}{3} = 1.2\text{ V/m}^2$$

25. Correct answer is [4.68].

At equilibrium

$$2T\sin\theta = mg$$
$$\Rightarrow \quad 2.\left(\frac{YA}{a}\right)x\sin\theta.\sin\theta = mg$$
$$\Rightarrow \quad \frac{2YA}{a}\,x.\,\frac{x^2}{a^2} = mg$$
$$\Rightarrow \quad x = \left\{\frac{a^3 mg}{2YA}\right\}^{\frac{1}{3}}$$
$$= \left\{\frac{1\text{ m}\times 5\text{ kg}\times 10\text{ m/s}^2}{2(2.4\times 10^9\text{ N/m}^2)\times 10^{-4}\text{ m}^2}\right\}^{\frac{1}{3}}$$
$$= 4.68\text{ cm}$$

Chemistry

26. Option (1) is correct.

Covalent radius of Atom $= \frac{1}{2}\times$ distance between two nuclei

$$r_{Br_2} = \frac{3.2}{2} = 1.6\,\text{Å}$$

27. Option (4) is correct.

$PCl_5 = sp^3d$, no lone pairs, Trigonal pyramidal
$F_2O = sp^3$, two lone pairs, V-shape
$BCl_3 = sp^2$, no lone pairs, Triangular planar
$NH_3 = sp^3$, one lone pair, Triangular pyramidal

28. Option (1) is correct.

Alkaline solution of Ni^{2+} reacts with dimethyl glyoxime to form red precipitate of nickel dimethylglyoxime.

$$2DMG + NiCl_2 + 2NH_4OH \rightarrow \underset{\text{(Red ppt)}}{[Ni(DMG)_2]\downarrow} + 2NH_4Cl + 2H_2O$$

29. Option (3) is correct.

$$W = -2.303\,nRT\log\frac{V_2}{V_1}$$
$$= -2.303\times 2\times 2\times 300\log\frac{50}{15}$$
$$= -2763.6\,(\log 10 - \log 3)$$
$$= -2763.6\,(1 - 0.4771)$$
$$= -2763.6\times 0.5229$$
$$= -1445\text{ cal.}$$

30. Option (2) is correct.

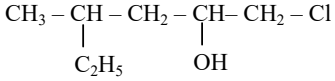
$$Bi_2S_3 \rightleftharpoons \underset{2s}{2Bi^{3+}} + \underset{3s}{3S^{2-}}$$
$$K_{sp} = (2s)^2\,(3s)^3 = 108\,s^5$$
$$108\times 10^{-75} = 108\,s^5$$
$$s = 1.0\times 10^{-15}\text{ mol/L}$$

31. Option (3) is correct.

For weak acid,

$$[H^+] = \sqrt{K_a\cdot C}$$
$$= \sqrt{0.1\times 2\times 10^{-5}}$$
$$= \sqrt{2\times 10^{-6}}$$
$$= \sqrt{2}\times 10^{-3}$$
$$pH = -\log[H^+]$$
$$= -\log\left(\sqrt{2}\times 10^{-3}\right)$$
$$pH = 3 - \log(1.414) \quad (\log 1.414 = 0.150)$$
$$= 2.85$$

32. Option (1) is correct.



1-Chloro-4 methyl hexan-2-ol

33. Option (1) is correct.

Statement I is correct. In asymmetrical molecule, the number of isomers can be calculated by using the formula $= 2^n$.

Where n = no. of double bond showing geometrical isomer.

Statement II is correct. To show geometrical isomerism, molecule must be plannar and have restricted rotation.

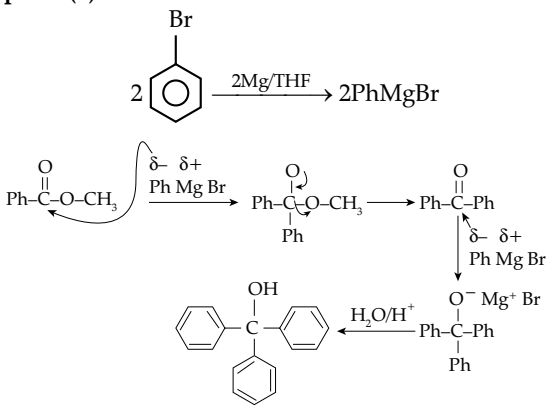
34. Option (1) is correct.

π–bond present in alkenes is weaker than the σ–bond present in alkanes, which makes alkenes less stable. Therefore, Statement I is correct. A carbon–carbon double bond is stronger than a carbon–carbon single bond because more energy

is required to break a sigma and pi bond than to break a sigma bond only. Therefore, statement II is also correct.

C≡C	120 pm
C=C	134 pm
C–C	154 pm

35. Option (2) is correct.



36. Option (2) is correct.

Cerium exists in two different oxidation state +3, +4

It shows Ce^{+4} and acts as a strong oxidising agent and accepts electron. Most stable and common oxidation state of lanthanoids is +3 because of the half filled *d* subshell.

37. Option (3) is correct.

$$[HCOOH] = 0.5 \text{ mol/L}$$

$\Rightarrow$ $(0.5\text{ml} \times 1.05\,\text{m/mL})$ HCOOH in 1 L

$\Rightarrow$ 0.525 g HCOOH in 1 L

$$m = \frac{(0.525/46)}{1 \text{ kg}} \text{ mol}$$

[Assuming dilute solution]

$$\therefore \quad \Delta T_f = iK_f m \Rightarrow i = \frac{\Delta T_f}{K_f m} = \frac{0.0405 \times 46}{1.86 \times 0.525} = 1.9$$

38. Option (2) is correct.

These are two main conditons for hyper conjugation:

(i) The presence of a hydrogen atom (C–H) adjacent to a sp^2 hybrid carbon (Carbocation).

(ii) The presence of a lone pair adjacent to a sp^2 hybrid carbon.

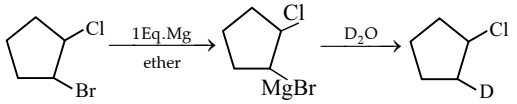
39. Option (3) is correct.

$$R - C \equiv CH \xrightarrow{Na/liq\,NH_3} R - C \equiv \bar{C}\ Na^+$$

$$\xrightarrow{CH_3Cl} R - C \equiv C - CH_3$$

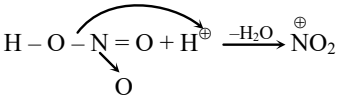
The end product of the reaction will be higher alkyne having one carbon more than 1-alkyne.

40. Option (1) is correct.



41. Option (3) is correct.

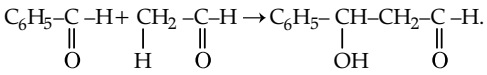
$$H_2SO_4 \rightarrow H^+ + HSO_4^-$$



42. Option (3) is correct.

Ethanol gives the iodoform test (heating with iodine and alkali) but methanol does not give this test.

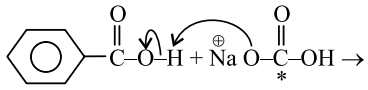
43. Option (4) is correct.



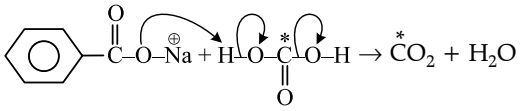
Aldol

In this reaction, only one kind of cross aldol is formed, because benzaldehyde does not have α – H.

44. Option (1) is correct.



Sod. bicarbonate



45. Option (3) is correct.

$C_2H_5NH_2 + HI \rightarrow$ C_2H_5NH–I

Ethyl ammonium iodide

$(CH_3)_3N + HBr \rightarrow$ $(CH_3)_3\overset{+}{N}HBr^-$

trimethyl ammonium bromide

46. Correct answer is [15.00].

Given,

$t_{1/2}$ of A = 5 min $\Rightarrow K = \lambda v^{2/5}$

$t_{1/2}$ of B = 15 min $\Rightarrow K = \lambda v^{2/15}$

Also, [A] = 4 [B]

For first order reaction;

$$\log R_o = \log R_t + kt$$

or $$R_t = R_o\, e^{-kt}$$

According to the question,

$$R_o\,(A) = 4\, R_o\,(B)$$

$$R_{A_t} = R_{B_t}$$

$$\frac{R_{A_{(t)}}}{R_{B_{(t)}}} = \frac{R_o(A) \times e^{-kt}}{R_o(B) \times e^{-kt}}$$

$$\frac{R_{A_{(t)}}}{R_{A_{(t)}}} = \frac{4R_{o_{(B)}} \times e^{-k_A t}}{R_{o_{(B)}} \times e^{-k_B t}}$$

$$\Rightarrow \quad 4 = e\left[\frac{\ln 2}{5} - \frac{\ln 2}{15}\right] \times t$$

Taking ln on both side,

$$\ln 4 = \left[\frac{\ln 2}{5} - \frac{\ln 2}{15}\right] t$$

$$\ln (2)^2 = \left[\frac{\ln 2}{5} - \frac{\ln 2}{15}\right] t$$

$$2 \ln 2 = \left[\frac{\ln 2}{5} - \frac{\ln 2}{15}\right] t$$

$$2 = \frac{2}{15} \times t$$

$$t = 15 \text{ min}$$

47. Correct answer is [0.34].

$$\Delta G^\circ = -nFE^\circ$$

$$E^\circ = \frac{-10.26 \times 1000 \times 4.18}{-2 \times 96500}$$

$$= 0.222$$

$$E = 0.222 - \frac{0.0591}{2} \log \frac{(0.1)^2 (0.1)^2}{1}$$

$$= 0.34 \text{ V}$$

48. Correct answer is [2].

$A = CH_3CHO$

$B = H—\overset{\overset{O}{\|}}{C}—ONa$

$C = CHI_3$

$D = HC \equiv CH$

49. Correct answer is [57].

From $\Delta H = \Delta U + \Delta n_g RT$

$\Delta U = -59.6$ kJ/mol

$T = 27°C = 27 + 273 = 300K$

$H_2F_{2(q)} \longrightarrow H_{2(g)} + F_{2(g)}$

$\Delta n_g = 2 - 1 = 1,$

$$R = 8.314 \times 10^{-3} \frac{kJ}{mol.K}$$

$$\Delta H = -59.6 + 1 \times 8.31 \times 10^{-3} \frac{kJ}{mol\ k} \times 300K$$

$\Delta H = -57.10$ kJ/mol

$\Delta H \approx -57$

50. Correct answer is [8].

(a) The number of unpaired e^- in $[FeF_6]^{3-}$ is 5.

Fe (z = 26) : [Ar] $3d^6$ $4s^2$

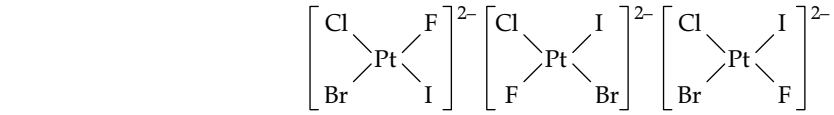
Fe^{3+} : [Ar] $3d^5$ $4s^0$

1	1	1	1	1

$F^\ominus$ ion is a weak field ligand which does not pair the unpaired e^-. So the total number of unpaired e^- = 5.

$$\boxed{x = 5}$$

(b) The number of geometrical isomers of [Pt Cl Br FI]$^{2-}$ is 3.



Total isomers are 3.

$$\boxed{y = 3}$$

(c) The oxidation state of central atom in Ni $(CO)_4$ is 0.

$$\boxed{z = 0}$$

The value of $x + y + z = 5 + 3 + 0 = 8$

Mathematics

51. Option (2) is correct.

Given, Relation R = {(a, b), (b, c)} on the set {a, b, c}

For symmetric $(b, a), (c, b) \in R$

and for transitive $(a, c) \in R$

Now, relation $R = \{(a, b), (b, c), (b, a), (c, b), (a, c)\}$

Again for symmetric $(c, a) \in R$

and for transitive

$\because \quad (a, b), (b, a) \in R$

$\therefore \quad (a, a) \in R$

and $(b, c), (c, b) \in R$

$\Rightarrow \quad (b, b) \in R$

and $(c, a), (a, c) \in R$

$\Rightarrow \quad (c, c) \in R$

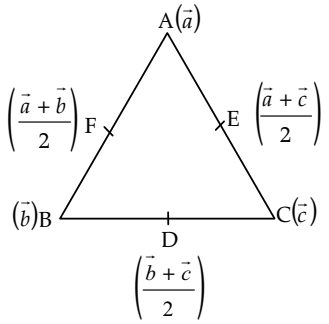
$\therefore$ 7 elements {(b, a), (c, b), (a, c), (c, a), (a, a), (b, b) (c, c)} must be added to the given relation, so that it becomes symmetric and transitive.

Hint:

(1) A relation R on a set A is said to be symmetric relation iff $(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b, \in A$.

(2) A relation R on a set A is said to be transitive relation iff $(a, b) \in R$, $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in A$.

52. Option (2) is correct.



$\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF}$

$= \left(\frac{\vec{b}+\vec{c}}{2} - \vec{a}\right) + \left(\frac{\vec{a}+\vec{c}}{2} - \vec{b}\right) + \left(\frac{\vec{a}+\vec{b}}{2} - \vec{c}\right)$

$= \left(\vec{a}+\vec{b}+\vec{c}\right) - \left(\vec{a}+\vec{b}+\vec{c}\right)$

$= 0$

Hence, $|\overrightarrow{AD} + \overrightarrow{BE} + \overrightarrow{CF}| = 0$

53. Option (3) is correct.

Given, $I_m = \int \frac{x^m}{y} dx$

$aI_3 + bI_2 + cI_1 = x^2y$

$\Rightarrow a\int \frac{x^3}{y}dx + b\int \frac{x^2}{y}dx + c\int \frac{x}{y}dx = x^2y$

$\Rightarrow \int \frac{ax^3+bx^2+cx}{y}dx = x^2y$

$\Rightarrow \frac{ax^3+bx^2+cx}{y} = \frac{d}{dx}(x^2y)$

$\Rightarrow \frac{ax^3+bx^2+cx}{y} = 2xy + x^2\frac{dy}{dx}$ ...(1)

Also, $y^2 = 2x^2 + 4x + 3$

$\Rightarrow 2y\frac{dy}{dx} = 4x + 4$

$\Rightarrow \frac{dy}{dx} = \frac{2x+2}{y}$ ...(2)

From eq. (1) and (2)

$\Rightarrow \frac{ax^3+bx^2+cx}{y} = 2xy + \frac{x^2}{y}(2x+2)$

$\Rightarrow ax^3 + bx^2 + cx = 2xy^2 + 2x^3 + 2x^2$

$\Rightarrow ax^3 + bx^2 + cx = 2x(2x^2 + 4x + 3) + 2x^3 + 2x^2$

$\Rightarrow ax^3 + bx^2 + cx = 4x^3 + 8x^2 + 6x + 2x^3 + 2x^2$

$\Rightarrow ax^3 + bx^2 + cx = 6x^3 + 10x^2 + 6x$

On comparing both the sides we get,

$a = 6, b = 10, c = 6$

$\therefore \quad {}^{a+c}P_{b+1} = {}^{12}P_{11} = \frac{12!}{1!} = 12!$

Hint:

(1) Use addition of integration and product rule of differentiation and simplify.

(2) ${}^nP_r = \frac{n!}{(n-r)!}$

54. Option (2) is correct.

Let $I = \int_0^\infty \left[\frac{2}{e^x}\right] dx$

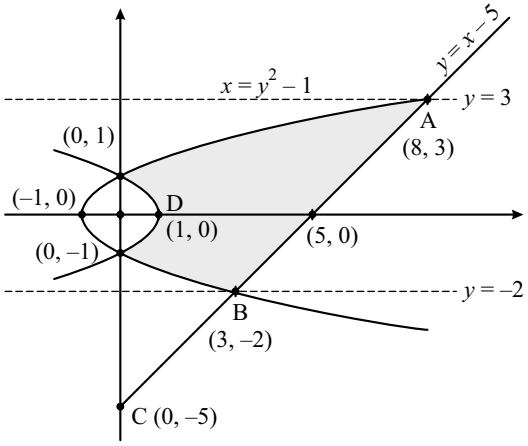
$= \int_0^{\log_e 2} dx + \int_{\log_e 2}^\infty 0 dx$

$= \log_e 2$

55. Option (1) is correct.

Given, Curves $x = |y^2 - 1|$ and $y = x - 5$

$\Rightarrow x = |y^2 - 1| = \begin{cases} -(y^2-1); & -1 < y < 1 \\ y^2 - 1; & y \le -1 \ \& \ y \ge 1 \end{cases}$



Let find intersection points of $y = x - 5$ and $x = y^2 - 1$

$\Rightarrow y + 5 = y^2 - 1$

$\Rightarrow y^2 - y - 6 = 0$

$\Rightarrow (y - 3)(y + 2) = 0$

$\Rightarrow y = 3, -2$

$\Rightarrow$ Intersecting points are (8, 3) and (3, – 2)

$\therefore$ Required area $A = \int_{-2}^{3} \{(y+5) - |y^2 - 1|\} dy$

$\Rightarrow A = \int_{-2}^{3} \{(y+5)dy - \left\{\int_{-2}^{-1}(y^2-1)dy + \int_{-1}^{1} -(y^2-1)dy + \int_1^3 (y^2-1)dy\right\}$

$\Rightarrow A = \left[\frac{y^2}{2} + 5y\right]_{-2}^{3} - \left\{\left[\frac{y^3}{3} - y\right]_{-2}^{-1}\right.$

$$-\left[\frac{y^3}{3}-y\right]_{-1}^{1}+\left[\frac{y^3}{3}-y\right]_{1}^{3}\Bigg\}$$

$$\Rightarrow \text{A} = \left[\frac{9}{2}+15-2+10\right]-\left[-\frac{1}{3}+1+\frac{8}{3}-2\right]$$

$$+\left[\frac{1}{3}-1+\frac{1}{3}-1\right]-\left[\frac{27}{3}-3-\frac{1}{3}+1\right]$$

$$\Rightarrow \text{A} = \frac{39}{2}+8-\frac{4}{3}-\frac{4}{3}-\frac{20}{3}$$

$$\Rightarrow \text{A} = \frac{55}{2}-\frac{28}{3}$$

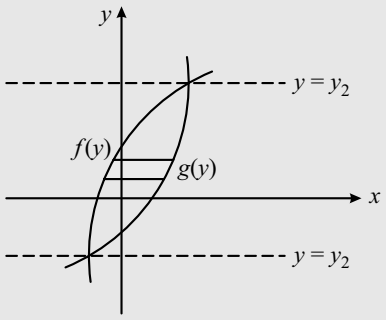
$$\Rightarrow \text{A} = \frac{109}{6} \text{ sq. units}$$

Hint:

(1) Draw the curves using the concept of modulus function and identity the required region.

(2) Area bounded by two curves $x = g(y)$ and $x = f(y)$ such that $g(y) > f(y)$ is given by

$$\Rightarrow \qquad \text{A} = \int_{y_1}^{y_2}[g(y)-f(y)]dy$$



56. Option (3) is correct.

$$\text{I} = \int\sqrt{1+x^{2n}}\left(\frac{\log_e(1+x^{2n})-2n\log_e x}{x^{3n+1}}\right)dx$$

$$\Rightarrow \text{I} = \int\sqrt{\frac{1}{x^{2n}}+1}\left\{\frac{\log_e\left(\frac{1}{x^{2n}}+1\right)}{x^{2n+1}}\right\}dx$$

$$\Rightarrow \text{Let} \qquad \frac{1}{x^{2n}}+1 = u^2$$

$$\Rightarrow \qquad -\frac{2n}{x^{2n+1}}dx = 2u\,du$$

$$\text{So,} \quad \text{I} = -\int\frac{u^2\log_e u^2}{n}du$$

$$\Rightarrow \text{I} = -\frac{2}{n}\int u^2\log_e u\,du$$

$$\Rightarrow \text{I} = -\frac{2}{n}\left[\log_e u\frac{u^3}{3}-\int\left(\frac{1}{u}\right)\frac{u^3}{3}du\right]$$

$$\Rightarrow \text{I} = -\frac{2}{n}\left[\frac{u^3}{3}\log_e u-\frac{1}{3}\,\frac{u^3}{3}\right]+c$$

$$\Rightarrow \text{I} = -\frac{2}{3n}u^3\left[\log_e u-\frac{1}{3}\right]+c$$

$$\Rightarrow \text{I} = -\frac{2}{3n}\left(\frac{1}{x^{2n}}+1\right)^{3/2}$$

$$\left[\log_e\left(1+\frac{1}{x^{2n}}\right)^{1/2}-\frac{1}{3}\right]+c$$

$$\Rightarrow \text{I} = -\frac{2}{3n}\frac{\sqrt{(1+x^{2n})^3}}{x^{3n}}$$

$$\left[\frac{1}{2}\log_e(1+x^{2n})-\frac{1}{2}\log_e x^{2n}-\frac{1}{3}\right]+c$$

Hint:

(1) Simplify given integral and substitute $\frac{1}{x^{2n}}+1=u^2$ and solved further using integration by parts method.

(2) $$\int f(x)g(x)dx = f(x)\int g(x)dx$$

$$-\int\left[f'(x)\left\{\int g(x)\,dx\right\}\right]dx$$

57. Option (2) is correct.

Given, $\quad y = a\log_e|x| + bx^2 + x$

$$\Rightarrow \qquad y = \begin{cases} a\log_e x+bx^2+x;\ x>0 \\ a\log_e(-x)+bx^2+x;\ x<0 \end{cases}$$

$$\Rightarrow \qquad \frac{dy}{dx} = \begin{cases} \frac{a}{x}+2bx+1,\ x>0 \\ \frac{a}{x}+2bx+1,\ x<0 \end{cases}$$

Since, $y = f(x)$ has it's extremum values at $x = -1$ and $x = 2$

$$\therefore \left(\frac{dy}{dx}\right)_{x=-1}=0 \text{ and } \left(\frac{dy}{dx}\right)_{x=2}=0$$

$\Rightarrow -a - 2b + 1 = 0$ and $\frac{a}{2} + 4b + 1 = 0$

$\Rightarrow \qquad 2b + a = 1$ and $a + 8b = -2$

$\Rightarrow \qquad a = 2$ and $b = -\frac{1}{2}$

58. Option (3) is correct.

$$\frac{dy}{dx} = \frac{a\cos\theta}{-a\sin\theta+\dfrac{a\sec^2\dfrac{\theta}{2}}{\tan\dfrac{\theta}{2}}\times\dfrac{1}{2}}$$

$$= \frac{a\cos\theta}{-a\sin\theta + \dfrac{a}{2\cos\dfrac{\theta}{2}\sin\dfrac{\theta}{2}}}$$

$$= \frac{a\cos\theta}{\dfrac{a}{\sin\theta} - a\sin\theta}$$

$$= \frac{a\cos\theta\sin\theta}{a - a\sin^2\theta}$$

$$= \frac{\cos\theta\sin\theta}{\cos^2\theta}$$

$$= \tan\theta$$

59. Option (3) is correct.

As we know,

$$0 \le \sin^2 x \le 1$$

$$\Rightarrow \quad 0 \le 16\sin^2 x \le 16$$

$$\Rightarrow \quad 1 \le 16\sin^2 x + 1 \le 17$$

$$\Rightarrow \quad \log_2(1) \le \log_2(16\sin^2 x + 1) \le \log_2(17)$$

$$\Rightarrow \quad 0 \le \log_2(16\sin^2 x + 1) \le \log_2(17)$$

$$\Rightarrow \quad 0 \ge -\log_2(16\sin^2 x + 1) \ge -\log_2(17)$$

$$\Rightarrow \quad 2 \ge 2 - \log_2(16\sin^2 x + 1) \ge 2 - \log_2(17)$$

As logarithm of negative value is not defined so we will consider

$$2 \ge 2 - \log_2(16\sin^2 x + 1) > 0$$

$$\Rightarrow \log_{\sqrt{2}} 2 =$$

$$\log_{\sqrt{2}}(2 - \log_2(16\sin^2 x + 1)) > \log\sqrt{2}$$

$$\Rightarrow \log_{\sqrt{2}}(\sqrt{2})^2 = \log_{\sqrt{2}}(2 - \log_2(16\sin^2 x + 1)) > -\infty$$

$$\Rightarrow \quad 2 \ge \log_{\sqrt{2}}(2 - \log_2(16\sin^2 x + 1)) > -\infty$$

Range of given function is $(-\infty, 2]$

$$\Rightarrow \quad P = 2$$

Hint: Use $0 \le \sin^2 x \le 1$ and then use properties of inequality and logarithm to solve further.

60. Option (2) is correct.

Given, $\lim\limits_{x\to 1} \sec^{-1}\left(\dfrac{k^2}{\log_e x} - \dfrac{k^2}{x-1}\right)$ exists.

Let us consider

$$= \lim_{x\to 1}\left(\frac{k^2}{\log_e x} - \frac{k^2}{x-1}\right)$$

$$= \lim_{x\to 1} k^2\left(\frac{1}{\log_e x} - \frac{1}{x-1}\right)$$

$$= \lim_{x\to 1} k^2\left(\frac{(x-1) - \log_e x}{(\log_e x)(x-1)}\right)$$

$$= \lim_{x\to 1} k^2\left(\frac{(x-1) - \log_e\{1 + (x-1)\}}{(x-1)\log_e\{1 + (x-1)\}}\right)$$

$$\lim_{x\to 1} k^2\left(\frac{(x-1) - (x-1) + \dfrac{(x-1)^2}{2} - \dfrac{(x-1)^3}{3} + \ldots.}{(x-1)\left\{(x-1) - \dfrac{(x-1)^2}{2} + \dfrac{(x-1)^3}{3} + \ldots..\right\}}\right)$$

$$= \lim_{x\to 1} k^2\left(\frac{(x-1)^2\left\{\dfrac{1}{2} - \dfrac{(x-1)}{3} + \ldots..\right\}}{(x-1)^2\left\{1 - \dfrac{(x-1)}{2} + \dfrac{(x-1)^2}{3}\ldots..\right\}}\right)$$

$$= \lim_{x\to 1} k^2\left(\frac{\dfrac{1}{2} - \dfrac{(x-1)}{3} + \ldots..}{1 - \dfrac{(x-1)}{2} + \ldots..}\right)$$

$$\Rightarrow \quad = \frac{k^2}{2}$$

Now, given that $\lim\limits_{x\to 1} \sec^{-1}\left(\dfrac{k^2}{2}\right)$ exists.

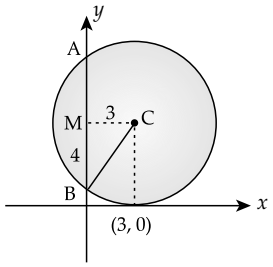
$$\Rightarrow \quad \frac{k^2}{2} \ge 1$$

$\Rightarrow k^2$ has smallest value as 2.

Hint: Use $\log_e(x+1) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \ldots..$ and simplify.

61. Option (1) is correct.

Let centre of circle is C and circle cuts the y-axis at B and A. Let mid-point of chord BA is M.



$$CB = \sqrt{MC^2 + MB^2}$$

$$\Rightarrow \quad CB = \sqrt{3^2 + 4^2}$$

$$= 5 = \text{radius of circle}$$

$\therefore$ Equation of circle is,

$(x-3)^2 + (y-5)^2 = 5^2$

(3, 10) satisfies this equation.

Although, there will be another circle satisfying the same conditions that will lie below the x-axis having equation

$(x-3)^2 + (y+5)^2 = 5^2$

62. Option (1) is correct.

Given, $\sum_{i=1}^{8}(x_i-7) = 8$

and $\sum_{i=1}^{8}(x_i-7)^2 = 40$

Let $X_i = x_i - 7$

Now, $\bar{X} = \frac{1}{8}\sum_{i=1}^{8}X_i$

$\Rightarrow \bar{X} = \frac{1}{8}\sum_{i=1}^{8}x_i - 7$

$\Rightarrow \bar{X} = \frac{1}{8}\times 8 = 1$

Now, Var (X) = $\sigma^2 = \frac{1}{N}\sum x^2 - (\bar{X})^2$

$= \frac{1}{8}\sum(x_i-7)^2 - (1)^2$

$= \frac{1}{8}\times 40 - 1$

$= 4$

$\because$ S.D. $= \sqrt{\text{Var (X)}}$

$\Rightarrow$ S.D. $= \sqrt{4} = 2$

As we know by property of S.D., it is independent of change of origin.

So, standard deviations of $x_1, x_2, x_3, \ldots\ldots x_8$ is 2.

So, both statements are true and statement 2 is correct explanation for statement-1.

Hint:

(1) Use Var $(x) = \sigma^2 = \frac{1}{N}\sum x^2 - (\bar{x})^2$

(2) S.D. $= \sqrt{\text{Var}(x)}$

(3) Recall the properties of standard deviation.

63. Option (3) is correct.

Given, A (1, 6, 3)

AB is perpendicular to

$x = \frac{y-1}{2} = \frac{z-2}{3}$

Let $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3} = \lambda$

$\Rightarrow$ Coordinates of B are $(\lambda, 2\lambda + 1, 3\lambda + 2)$

Now direction ratios of AB are $\lambda - 1, 2\lambda + 1 - 6, 3\lambda + 2 - 3$

$\Rightarrow$ D.Rs of AB are $\lambda - 1, 2\lambda - 5, 3\lambda - 1$

Also, D.Rs of the given line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ are 1, 2, 3.

Now, AB is perpendicular to the given line.

$\Rightarrow (\lambda - 1)1 + (2\lambda - 5)2 + (3\lambda - 1)3 = 0$

$\Rightarrow \lambda - 1 + 4\lambda - 10 + 9\lambda - 3 = 0$

$\Rightarrow 14\lambda - 14 = 0$

$\Rightarrow \lambda = 1$

$\Rightarrow$ Coordinates of B are (1, 2(1) + 1, 3(1) + 2)

i.e., (1, 3, 5)

Let the image of point A (1, 6, 3) be A′ (x_1, y_1, z_1)

$\Rightarrow$ B will be the mid point of AA′

$\therefore$ By mid point formula,

$\frac{x_1+1}{2} = 1$

$\frac{y_1+6}{2} = 3$

$\frac{z_1+3}{2} = 5$

$\Rightarrow x_1 = 2 - 1$

$\Rightarrow x_1 = 1$

$\Rightarrow y_1 = 6 - 6$

$\Rightarrow y_1 = 0$

$\Rightarrow z_1 = 10 - 3$

$\Rightarrow z_1 = 7$

So, image of point A is (1, 0, 7) w.r.t. the given line.

As we know, projection of line segment joining P (x, y, z) and Q (x_2, y_2, z_2) on a line having direction cosines l, m, n is.

$|l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)|$

$\therefore$ Projection of AB with A (1, 6, 3) and B (1, 3, 5) on line having direction cosines $\frac{1}{\sqrt{2}}, 0, -\frac{1}{\sqrt{2}}$ is

$\Rightarrow \left|\frac{1}{\sqrt{2}}(1-1) + 0(6-3) + \left(-\frac{1}{\sqrt{2}}\right)(3-5)\right|$

$= \left|0 + 0 + \sqrt{2}\right|$

$= \sqrt{2}$

Also, by distance formula

AB $= \sqrt{(1-1)^2 + (6-3)^2 + (3-5)^2}$

$\Rightarrow$ AB $= \sqrt{0 + 9 + 4}$

$\Rightarrow$ AB $= \sqrt{13}$

Hint:

(1) If direction ratios of two perpendicular lines are a_1, b_1, c_1 and a_2, b_2, c_2 then $a_1a_2 + b_1b_2 + c_1c_2 = 0$.

(2) The projection of line segment joining P (x_1, y_1, z_1) and (x_2, y_2, z_2) on a line having direction cosines l, m, n is

$|l(x_2 - x_1) + m(y_2 - y_1) + n(z_2 - z_1)|$

64. Option (4) is correct.

Equation of ellipse

$$\frac{x^2}{16} + \frac{y^2}{7} = 1 \quad \text{....(i)}$$

And equation of hyperbola

$$\frac{x^2}{144} - \frac{y^2}{\alpha} = \frac{1}{25} \quad \text{....(ii)}$$

$$\Rightarrow \quad \frac{x^2}{\frac{144}{25}} - \frac{y^2}{\frac{\alpha}{25}} = 1$$

Have same foci, then

$$a^2 - b^2 = l^2 + m^2$$

$$\Rightarrow \quad 16 - 7 = \frac{144}{25} + \frac{\alpha}{25}$$

$$\Rightarrow \quad 9 = \frac{144}{25} + \frac{\alpha}{25}$$

$$\Rightarrow \quad 9 \times 25 = 144 + \alpha$$

$$\Rightarrow \quad \alpha = 225 - 144 = 81$$

$$\therefore \Rightarrow \quad \alpha = 81$$

Latus Rectum of hyperbola

$$= \frac{2b^2}{a} = \frac{2\times\frac{\alpha}{25}}{\frac{12}{5}}$$

$$= \frac{2\times 81}{25}\times\frac{5}{12} = \frac{27}{10}$$

65. Option (3) is correct.

Let roots be α and β then from equation

$$-x^2 - bx + a = 0$$

We have, $\alpha\beta = -a < 0$

$\Rightarrow$ Both the roots are in opposite sign and

$$\alpha + \beta = -b < 0$$

$\Rightarrow$ Both roots are in opposite sign and greater root in magnitude is negative.

66. Option (2) is correct.

Given, system of linear equations

$$-x + y + 2z = 0$$
$$3x - ay + 5z = 1$$
$$2x - 2y - az = 7$$

$$\Delta = \begin{vmatrix} -1 & 1 & 2 \\ 3 & -a & 5 \\ 2 & -2 & -a \end{vmatrix}$$

$\Rightarrow \quad \Delta = -1(a^2 + 10) - 1(-3a - 10) + 2(-6 + 2a)$

$\Rightarrow \quad \Delta = -a^2 - 10 + 3a + 10 - 12 + 4a$

$\Rightarrow \quad \Delta = -a^2 + 7a - 12$

$\Rightarrow \quad \Delta = -(a - 3)(a - 4)$

Also, $\Delta_1 = \begin{vmatrix} 0 & 1 & 2 \\ 1 & -a & 5 \\ 7 & -2 & -a \end{vmatrix}$

$\Rightarrow \quad \Delta_1 = 0 - 1(-a - 35) + 2(-2 + 7a)$

$\Rightarrow \quad \Delta_1 = a + 35 - 4 + 14a$

$\Rightarrow \quad \Delta_1 = 15a + 31$

For an inconsistent system $\Delta = 0$ and $\Delta_1 \neq 0$

$\Rightarrow \quad -(a - 3)(a - 4) = 0$

$\Rightarrow \quad a = 3$ or 4

and for $a = 3$, $\Delta_1 = 45 + 31 = 76 \neq 0$.

for $a = 4$, $\Delta_1 = 60 + 31 = 91 \neq 0$.

$\Rightarrow \quad P = \{3, 4\}$

$\Rightarrow \quad \Sigma P = 7$

For infinitely many solutions, $\Delta = 0$ and $\Delta_1 = \Delta_2 = \Delta_3 = 0$.

$\Rightarrow \quad -(a - 3)(a - 4) = 0$

$\Rightarrow \quad a = 3$ or 4

for $a = 3$, $\Delta_1 = 76 \neq 0$

for $a = 4$, $\Delta_1 = 91 \neq 0$

So, no values of a possible

$\Rightarrow \quad Q = \{\phi\}$

$\Rightarrow \quad \Sigma Q = 0$

$\Rightarrow \quad \Sigma P + \Sigma Q = 7 + 0 = 7.$

Hint:

(1) For an inconsistent system $\Delta = 0$ and one of the $\Delta_1, \Delta_2, \Delta_3$ not equals to zero.

(2) For infinitely many solution $\Delta = 0$ and $\Delta_1 = \Delta_2 = \Delta_3 = 0$.

67. Option (1) is correct.

$\because$ Sum of n terms of A.P with first terms a and common difference d is given by

$$S_n = \frac{n}{2}[a + (n-1)d]$$

$$\therefore \quad \frac{S_{n_1}}{S_{n_2}} = \frac{\frac{n}{2}\left[2a_1 + (n-1)d_1\right]}{\frac{n}{2}\left[2b_1 + (n-1)d_2\right]} = \frac{2n}{n+1}$$

$$\Rightarrow \quad \frac{a_1 + \frac{(n-1)}{2}d_1}{b_1 + \frac{(n-1)}{2}d_2} = \frac{2n}{n+1} \quad \text{...(i)}$$

For T_8, we know $\frac{n-1}{2} = 7$

$\Rightarrow \quad n = 15$

Put $n = 15$ in (i), we get

$$\frac{(T_8)_1}{(T_8)_2} = \frac{30}{16} = \frac{15}{8}$$

68. Option (2) is correct.

$$T_4 = {}^nC_3(ax)^{n-3}\left(\frac{1}{x}\right)^3$$

$$= {}^nC_3\, a^{n-3}\, x^{n-6} = \frac{5}{2} \text{ (given)}$$

$$\therefore \quad n - 6 = 0$$

$\Rightarrow \quad n = 6$

$\Rightarrow \quad T_4 = {}^6C_3 a^{6-3} = \frac{5}{2}$

$\Rightarrow \quad a^3 = \frac{1}{8} = \left(\frac{1}{2}\right)^3$

$\Rightarrow \quad a = \frac{1}{2}$

$\Rightarrow \quad a = \frac{1}{2},\ n = 6$

69. Option (1) is correct.

$${}^{(n+1)}C_{(n-2)} - {}^{(n+1)}C_{(n-1)} \le 100$$

$$\Rightarrow \frac{(n+1)!}{(n-2)!(n+1-n+2)!} - \frac{(n+1)!}{(n-1)!(n+1-n+1)!} \le 100$$

$$\Rightarrow \frac{(n+1)!}{(n-2)!3!} - \frac{(n+1)!}{(n-1)!2!} \le 100$$

$$\Rightarrow (n+1)!\left[\frac{1}{3!(n-2)!} - \frac{1}{2!(n-1)!}\right] \le 100$$

$$\Rightarrow (n+1)!\left[\frac{(n-1)! - 3(n-2)!}{6(n-1)!(n-2)!}\right] \le 100$$

$$\Rightarrow \frac{(n+1)n(n-1)!\left[(n-1)(n-2)! - 3(n-2)!\right]}{6(n-1)!(n-2)!} \le 100$$

$$\Rightarrow \frac{(n+1)\,n\,(n-1)!\,(n-2)!\,(n-1-3)}{6(n-1)!(n-2)!} \le 100$$

$$\Rightarrow \frac{(n+1)\,n\,(n-4)}{6} \le 100$$

$$\Rightarrow n(n+1)(n-4) \le 600$$

Put $n = 1, 2, 3, 4, 5, 6, 7, 8, 9, \ldots\ldots$

For $n = 5$, $600 \ge 5 \times 6 \times 1$ holds good

For $n = 8$, $600 \ge 8 \times 9 \times 4$
$= 288$ holds good

For $n = 9$, $600 \ge 9 \times 10 \times 5$
$= 450$ holds good

For $n = 10$, $600 \le 10 \times 11 \times 6$
$= 650$ does not hold

Hence, the number of positive integer which is satisfying the given inequality must be 9.

Shortcut Method:

$${}^{n+1}C_{n-2} - {}^{n+1}C_{n-1} \le 100$$

$$\Rightarrow \quad {}^{n+1}C_3 - {}^{n+1}C_2 \le 100$$

$$\Rightarrow \quad \frac{(n+1)n(n-1)}{6} - \frac{(n+1)n}{2} \le 100$$

$$\Rightarrow \quad n\,(n+1)\,(n-4) \le 600$$

Consider

$$f(x) = x\,(x+1)\,(x-4)\ ;\ x \ge 1$$
$$= x^3 - 3x^2 - 4x$$

$\Rightarrow \quad f'(x) = 3x^2 - 6x - 4$

If $\quad f'(x) > 0$

$$x \in \left(1 + \frac{\sqrt{21}}{3}, \infty\right)$$

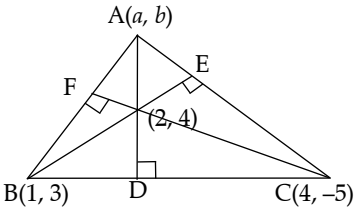
It is easy to check that

$f(10) = 10 \times 11 \times 6 = 660 > 600$

and $\quad f(9) = 9 \times 10 \times 5 = 450 < 600$

$\therefore \quad x = 9$ is the last integer so that $f(x) < 600$.

70. Option (3) is correct.



From the figure

Slope of $BE = 1$

Slope of $AC = -1$

$\because$ (slope of $BE \times$ slope of $AC = -1$)

$\therefore$ Equation AC is $x + y = -1$ ...(i)

Slope of $CF = \frac{-9}{2}$

Slope of $AB = \frac{2}{9}$

$\because$ (slope of $CF \times$ Slope of $AB = -1$)

$\therefore$ Equation AB is $2x - 9y = -25$ ...(ii)

Solving (i) and (ii), we get

$$(a, b) = \left(\frac{-34}{11}, \frac{23}{11}\right)$$

$\therefore$ The value of $33b + 22a = 1$

71. Correct answer is [19].

Given lines L_1: $\frac{x+3}{5} = \frac{y-1}{2} = \frac{z+4}{3} = \lambda$ (let)

L_2: $\frac{x-1}{1} = \frac{y+2}{3} = \frac{z-2}{-2} = \mu$ (let)

and line with direction ratios $\langle 0, 2, -1\rangle$ meets the line L_1 and L_2 at P and Q respectively.

Let the coordinates of P be $(5\lambda - 3, 2\lambda + 1, 3\lambda - 4)$ and coordinates of Q be $(\mu + 1, 3\mu - 2, -2\mu + 2)$

Now, direction ratios of line PQ

$= \mu - 5\lambda + 4, 3\mu - 2\lambda - 3,$
$-2\mu - 3\lambda + 6.$

$\therefore$ direction ratios of line PQ = 0, 2, –1

$\therefore \quad \frac{\mu-5\lambda+4}{0} = \frac{3\mu-2\lambda-3}{2} = \frac{-2\mu-3\lambda+6}{-1} = t$

$\Rightarrow \quad \mu - 5\lambda + 4 = 0$ ...(i)

$3\mu - 2\lambda - 3 = 2t$ ...(ii)

$-2\mu - 3\lambda + 6 = -t$ ...(iii)

From eq. (ii) and eq. (iii),

$3\mu - 2\lambda - 3 - 4\mu - 6\lambda + 12 = 0$

$\Rightarrow \quad -\mu - 8\lambda + 9 = 0$

$\Rightarrow \quad \mu + 8\lambda = 9$ ...(iv)

On solving eq. (i) and eq. (iv), $\mu = 1$ and $\lambda = 1$

$\therefore$ P = (2, 3, –1) and Q = (2, 1, 0)

Now,

$$\text{area of } \Delta OPQ = \frac{1}{2}|\overrightarrow{OP} \times \overrightarrow{OQ}|$$

$$\Rightarrow \quad \frac{1}{n}\sqrt{m} = \frac{1}{2}|(2\hat{i}+3\hat{j}-\hat{k})\times(2\hat{i}+\hat{j})|$$

$$\Rightarrow \quad \frac{1}{n}\sqrt{m} = \frac{1}{2}|\hat{i}-2\hat{j}-4\hat{k}|$$

$$\Rightarrow \quad \frac{1}{n}\sqrt{m} = \frac{1}{2}\sqrt{1+4+16}$$

$$\Rightarrow \quad \frac{1}{n}\sqrt{m} = \frac{1}{2}\sqrt{21}$$

$\Rightarrow n = 2$ and $m = 21$

$\Rightarrow \quad m - n = 19$

Hint:

(1) Assume coordinates of P and Q in parametric form and find direction ratios of PQ and compare it with given direction ratios of line and solved further.

(2) Area of triangle OPQ $= \frac{1}{2}|\overrightarrow{OP} \times \overrightarrow{OQ}|$

72. Correct answer is [17].

Required probability = when no machine has fault + when only one machine has fault + when only two machines have fault.

$$= {}^5C_0\left(\frac{3}{4}\right)^5 + {}^5C_1\left(\frac{1}{4}\right)\left(\frac{3}{4}\right)^4 + {}^5C_2\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^3$$

$$= \frac{243}{1024} + \frac{405}{1024} + \frac{270}{1024}$$

$$= \frac{918}{1024} = \frac{459}{512}$$

$$= \frac{27\times17}{64\times8}$$

$$\Rightarrow \quad \left(\frac{3}{4}\right)^3 \times k = \left(\frac{3}{4}\right)^3 \times \frac{17}{8}$$

$$\therefore \quad k = \frac{17}{8}$$

So, $8k = 17$

73. Correct answer is [78].

Given, $m = \int_{-1}^{10} \text{sgn}(x-[x])dx$

$\Rightarrow \quad m = \int_{-1}^{10} \text{sgn}\{x\}dx \quad \{\because x = [x] + \{x\}\}$

$\Rightarrow \quad m = 11\int_0^1 \text{sgn}(x)dx \quad \{\because$ Period of $\{x\}$ is 1$\}$

$\Rightarrow \quad m = 11\int_0^1 1\,dx$

$\Rightarrow \quad m = 11[x]_0^1$

$\Rightarrow \quad m = 11$

and $\quad n = \int_0^{26} \frac{e^{\cos\pi\{x\}}}{e^{\cos\pi\{x\}} + e^{-\cos\pi\{x\}}}dx$

$\Rightarrow \quad n = 26\int_0^1 \frac{e^{\cos\pi x}}{e^{\cos\pi x} + e^{-\cos\pi x}}dx$ ...(i)

$\Rightarrow \quad n = 26\int_0^1 \frac{e^{\cos\pi(1-x)}}{e^{\cos\pi(1-x)} + e^{-\cos\pi(1-x)}}dx$

$\left\{\because \int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right\}$

$\Rightarrow \quad n = 26\int_0^1 \frac{e^{-\cos\pi x}}{e^{-\cos\pi x} + e^{\cos\pi x}}dx$ ...(ii)

Adding eq. (i) and eq. (ii), we get,

$2n = 26\int_0^1 1.dx$

$\Rightarrow \quad 2n = 26[x]_0^1$

$\Rightarrow \quad n = 13$

$\therefore \quad {}^nC_m = {}^{13}C_{11} = {}^{13}C_2$

$= \frac{13.12}{2.1} = 78$

Hint:

(1) Use $x = [x] + \{x\}$

(2) $\text{sgn } x = \begin{cases} 1 & ; \quad x > 0 \\ -1 & ; \quad x < 0 \end{cases}$

(3) Use $\int_0^{nT} f(x)dx = n\int_0^T f(x)dx$;

where $f(x)$ is periodic function with period T

(4) $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

74. Correct answer is [600].

Since, D is a skew symmetric matrix

$\therefore \quad D + O^T = 0$

$\Rightarrow \quad a\alpha^2 + b\alpha + c = 0,$

$a\beta^2 + b\beta + c = 0$

$a\gamma^2 + b\gamma + c = 0$

$\Rightarrow \quad ax^2 + bx + c = 0$ has three roots

$\Rightarrow \quad a = b = c = 0$

$$\therefore \quad \lambda = \begin{vmatrix} 1 & 1 & 2 \\ 3 & 4 & 1 \\ 9 & 0 & 3 \end{vmatrix}$$

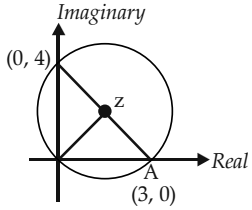
$$= -60$$

$$\Rightarrow \quad |10\lambda| = |10 \times -60| = 600$$

75. Correct answer is [5.00].

z is equidistant from (0, 0), (3, 0) and (0, 4)

$\Rightarrow z$ is at the circumcentre of ΔOAB



$$\Rightarrow \quad |z| = \frac{5}{2}$$

$$\Rightarrow \quad |2z| = 5.00$$

❐❐